

1. In this assessment, you will be tested on all of the different topics you have in covered this module. Good luck!

1 point

What is the derivative of the function $f(x) = x^{3/2} + \pi x^2 + \sqrt{7}$ evaluated at the point $x = 2$?

- ☐ $f'(2) = \frac{3}{2} + 4\pi$
☐ $f'(2) = \frac{3\sqrt{2}}{2} + 4\pi + \sqrt{7}$
☒ $f'(2) = \frac{3\sqrt{2}}{2} + 4\pi$
☐ $f'(2) = \frac{3}{2} + 4\pi + \sqrt{7}$

2. What is the derivative of the function $f(x) = x^3 \cos(x) e^x$?

1 point

- ☐ $f'(x) = -e^x x^3 \sin(x) + e^x x^3 \cos(x) + e^x x^2 \cos(x)$
☒ $f'(x) = -e^x x^3 \sin(x) + e^x x^3 \cos(x) + 3e^x x^2 \cos(x)$
☐ $f'(x) = -x^3 \sin(x) + e^x x^3 + 3e^x x^2 \cos(x)$
☐ $f'(x) = -3x^2 \sin(x) e^x$

3. What is the derivative of the function $f(x) = e^{[(x+1)^2]}$?

1 point

- ☐ $f'(x) = e^{[(x+1)^2]}$
☐ $f'(x) = e^{2(x+1)}$
☒ $f'(x) = 2(x+1)e^{[(x+1)^2]}$
☐ $f'(x) = (x+1)e^{[(x+1)^2]}$

4. What is the derivative of the function $f(x) = x^2 \cos(x^3)$?

1 point

- ☐ $f'(x) = 2x \sin(x^3) - 3x^4 \cos(x^3)$
☐ $f'(x) = 2x \cos(x^3) - 3x^4 \cos(x^3)$
☒ $f'(x) = 2x \cos(x^3) - 3x^4 \sin(x^3)$
☐ $f'(x) = 2x \sin(x^3) - 3x^4 \sin(x^3)$

5. What is the derivative of the function $f(x) = \sin(x) e^{\cos(x)}$ at the point $x = \pi$?

1 point

- ☒ $f'(\pi) = -\frac{1}{e}$
☐ $f'(\pi) = \frac{1}{e^2}$
☐ $f'(\pi) = -\frac{1}{e^2}$
☐ $f'(\pi) = \frac{1}{e}$